

PI Control for Optimal Spectral Abscissa in General Non-Minimum Phase Systems with Time Delay

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Abstract—In this study, we investigate the stabilization of general linear second-order non-minimum phase time-delay systems using PI controllers. Our primary objective is to determine optimal control gains that shift the spectral abscissa of the system as far left as possible, ensuring enhanced stability. To achieve this, we employ a comprehensive geometrical analysis, providing a methodology to tune the control gains that yield the desired decay rate. The presented approach not only contributes theoretical insights but also facilitates practical implementation. The effectiveness of the proposed methodology is demonstrated through numerical examples.

Index Terms—non-minimum phase, time-delay, PI control, frequency-domain

I. INTRODUCTION

Controlling second-order dynamics often involves addressing two common challenges: the presence of unstable zeros, commonly referred to as non-minimum phase (NPM), which produces limitations on the control of the system, as well as an initial response in the opposite direction of the steady state one, and the presence of input/output delays, which arise from a wide arrange of sources such as communication time, control decision, among others [1]–[4]. Both phenomena pose unique difficulties in terms of stabilization analysis, and their resolution holds significant industrial importance, given that second-order non-minimum phase plus time-delay systems capture the dynamics of a wide array of industrial processes, such as bio-chemical reactors, hydraulic turbines, airplanes, power electronics converters, among others [5]–[8].

The stabilization of time-delay systems has been extensively explored in the literature over the past few decades [9]–[12], with Proportional-Derivative-Integral (PID) controller strategies emerging as effective solutions in numerous cases [13]–[17]. It is noteworthy that, despite being a subject of in-depth study in control since more than a century ago [18], [19], PID controllers still present unresolved issues. Recent discussions in the literature have particularly focused on the improperly-posed nature of the implementation of the derivative action [20]–[22], a challenge that can often be addressed through the

incorporation of low-pass filters, among other methodologies. However, in scenarios involving unstable zeros, the use of low-pass filters cannot always solve the problem [23], and thus, the necessity to develop alternative solutions.

In light of these considerations, this paper opts to drop the derivative action, focusing instead on pure Proportional-Integral (PI) controllers. This decision is made to address the challenges posed by the presence of unstable zeros and time-delays, acknowledging that the improperly-posedness associated with the derivative action requires distinct considerations. It is worth mentioning that in the past there has been efforts to deal with second-order systems that present both unstable zeros, and input-output delays (see, for example [3] and [4]). However, there are two main downsides on such studies. On the one hand, it is common to replace the exponential term that emerges from the presence of the delay by a Padé approximation, which implies that there is information on the solutions that is being lost, which can lead to miss-tuning the controller as the stable region might change. On the other hand, in order to gain one extra *degree of freedom* it is common to consider a full PID scheme but, as already mentioned, the inclusion of derivative action requires solving problems that, to the best of the authors knowledge, remain open.

The contribution of this work is twofold. On the one hand, we study the geometry of the stable region in the parameter-space (k_p, k_i) for a non-minimum phase second-order delay system, by means of the $\mathcal{D}$ -partition theory which allow us to *partition* the parameter-space into regions characterized by having the same number of unstable roots. On the other hand, when the system can be stabilized, we propose an analytical method (for the stable open-loop case) and a geometrical methodology (for both the marginally stable and unstable open-loop cases) to find the controller gains that guarantee that the systems spectral abscissa is located as far left as possible.

Notations: Throughout this paper, the following standard notations are used: the set of real numbers is denoted by $\mathbb{R}$. In particular, $\mathbb{R}_+ := \{x \in \mathbb{R} : x \geq 0\}$, and for any given set $\mathcal{A}$, $\mathcal{A}^* := \mathcal{A} \setminus \{0\}$. Next, $\mathbb{C}$ ($\mathbb{C}_+$, $\mathbb{C}_-$) represents the set of complex numbers (with strictly positive/negative real parts) and $i := \sqrt{-1}$. For a complex number $z \in \mathbb{C}$, $\bar{z}$ represents its complex conjugate. For a real number x , $\text{sign}(x)$ denotes its sign, with $\text{sign}(x) \in \{0, \pm 1\}$. Finally, for a given set $\mathcal{A}$, its cardinality is denoted by $\text{card}\{\mathcal{A}\}$.

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II. PRELIMINARIES, PROBLEMS FORMULATION AND PREREQUISITES

We consider second-order linear time-invariant (LTI) single-input/single-output (SISO) systems with transfer function representation of the following form:

$$H_{yu}(s; h) := \frac{P(s)}{Q(s)} e^{-hs}, \quad (1)$$

where $h \in \mathbb{R}_+^*$ is an input delay, $P(s)$ and $Q(s)$ are polynomials in the complex variable s with real parameters given by:

$$P(s) = s - z, \quad (2)$$

and

$$Q(s) = s^2 + as + b, \quad (3)$$

where $s = z$ is an unstable zero with $z > 0$ making the system NMP, $(a, b) \in \mathbb{R}^2$, and $Q(z) \neq 0$. We consider the system (1) in closed-loop with a classical PI controller:

$$K(s) = k_p + \frac{k_i}{s}. \quad (4)$$

Considering a classical unity feedback, the closed-loop system characteristic function $\Delta : \mathbb{C} \times \mathbb{R}_+ \rightarrow \mathbb{C}$ is given by:

$$\Delta(s; h) := s^3 + 2as^2 + bs + e^{-hs}(k_i + k_p s)(s - z), \quad (5)$$

which is a quasipolynomial due to the presence of the exponential term.

A. Problem formulation

We introduce the following notions:

Definition 1 (Spectral abscissa [24]). *For the closed-loop system (1)-(4) with associated characteristic function $\Delta(s; h)$ given by (5), its spectral abscissa is defined as:*

$$\rho(\Delta(s; h)) = \max \{ \Re(s_0) : s_0 \in \mathbb{C}, \Delta(s_0; h) = 0 \}$$

Definition 2 (σ -stability). *The closed-loop system (1)-(4) is said to be σ -stable if its spectral abscissa $\rho(\Delta(s; h)) \leq -\sigma$, where $\sigma \in \mathbb{R}_+$.*

The problem addressed on this note is the following:

Problem 1. *For a second-order LTI SISO system with the transfer function given by (1) in closed-loop with a PI controller given by (4), find the gains $(\bar{k}_p, \bar{k}_i) \in \mathbb{R}^2$ such that the spectral abscissa of the closed-loop system achieves its minimum value.*

B. Stability crossing curves

Remark 1. *The characteristic roots of the closed-loop system (1)-(4) given by the zeros of the characteristic quasipolynomial $\Delta(s; h)$ given by (5) are continuous functions of its parameters [24], therefore a characteristic root can only cross from the left-half plane to the right-half plane (stability to instability), or from the right-half plane to the left-half plane (instability to stability), by crossing the imaginary axis.*

Consider the following definition:

Definition 3 (Frequency crossing set). *The frequency crossing set $\Omega \subset \mathbb{R}$ is the set of all ω such that, for a given h , there exist at least one pair of control gains $(k_p, k_i) \in \mathbb{R}^2$ such that*

$$\Delta(i\omega; h) = 0. \quad (6)$$

It is easy to observe that $\Delta(i\omega; h) = \overline{\Delta(-i\omega; h)}$, therefore in the following we consider only $\Omega \subset \mathbb{R}_+$.

Definition 4 (Stability crossing curves). *The stability crossing curve $\mathcal{K}$ is the set of all pairs $(k_p, k_i) \in \mathbb{R}^2$ for which there exist $\omega \in \Omega$. Any point $\vec{k} \in \mathcal{K}$ is referred to as a crossing point.*

III. MAIN RESULTS

A. Decomposition of the (k_p, k_i) plane

Proposition 1 (Crossing points characterization). *Consider the system described by the transfer function $H_{yu}(s; h)$ in (1) in closed-loop with the PI controller in (4). Then, the crossing points $\vec{k} = (k_p(\omega), k_i(\omega)) \in \mathbb{R}^2$ are given by:*

$$k_p(\omega) = \frac{\alpha_1 \cos(h\omega) + \alpha_2 \sin(h\omega)}{\omega^2 + z^2}, \quad (7)$$

$$k_i(\omega) = \frac{\omega(\alpha_3 \cos(h\omega) + \alpha_4 \sin(h\omega))}{\omega^2 + z^2}, \quad (8)$$

where:

$$\begin{aligned} \alpha_1 &:= bz - \omega^2(a + z), & \alpha_2 &:= \omega(-az - b + \omega^2), \\ \alpha_3 &:= \omega(\omega^2 - az - b), & \alpha_4 &:= \omega^2(a + z) - bz. \end{aligned}$$

Moreover, if $\omega = 0$, we have $k_i = 0$.

Sketch of proof. The proof follows straightforwardly by noting that if $\Delta(i\omega; h) = 0$ then $\Re\{\Delta(i\omega; h)\} = \Im\{\Delta(i\omega; h)\} = 0$. $\square$

Corollary 1. *Consider the system described by the transfer function $H_{yu}(s; h)$ in (1) in closed-loop with the PI controller in (4), then the stability crossing curve $\mathcal{K} = \mathcal{K}_\omega \cup \mathcal{K}_0$ is given by:*

$$\mathcal{K}_\omega := \left\{ \vec{k} \in \mathbb{R}^2 : \vec{k} := (k_p(\omega), k_i(\omega)), \forall \omega \in \Omega \right\},$$

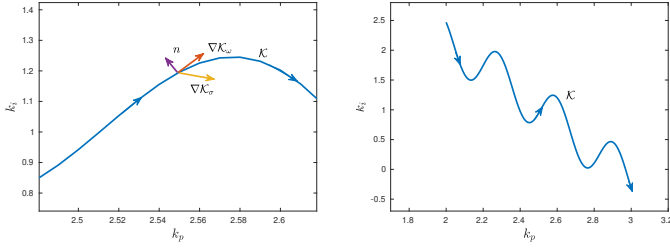
$$\mathcal{K}_0 := \left\{ \vec{k} \in \mathbb{R}^2 : \vec{k} := (k_p, 0), k_p \in \mathbb{R} \right\}.$$

It is worthy of mention that the crossing curve $\mathcal{K}$ decompose the parameter space into disjoint regions characterized by the same number of unstable roots. We refer to the region, if any, for which there are no unstable roots as the *stable region*, denoted by $\mathcal{S}$.

B. Crossing directions

In the following we refer to the direction of the curve $\mathcal{K}$ that correspond to increasing ω as the *positive direction* of the curve (see Fig. 1b). We have the following result:

Proposition 2. *Consider the system described by the transfer function H_{yu} in (1) in closed-loop with the PI controller in (4). Then, a pair of characteristic roots cross from $\mathbb{C}_+$ to $\mathbb{C}_-$*



(a) Geometrical interpretation of the crossing direction analysis. (b) Positive direction of the curve $\mathcal{K}$.

Fig. 1

if $\vec{k}$ traverses the stability crossing curve $\mathcal{K}$ from left to right with respect to the positive direction of the curve.

Sketch of proof. The proof follows similar steps to those presented in O. Diekmann *et al* [25] (Chapter XI). $\square$

C. The σ -stability crossing curves

Bearing Definition 2 in mind, we are interested in finding the stability crossing curves $\mathcal{K}_\sigma$ for a given $\sigma \in \mathbb{R}_+$. Studying the σ -stability is therefore equivalent to considering the change of variable $s \rightarrow s - \sigma$ and rewriting $\Delta(s; h)$ in (5) as $\Delta(s - \sigma; h)$. We have the following result:

Proposition 3 (σ -stability crossing points characterization). *Let $\sigma \in \mathbb{R}_+$ and $\omega \in \Omega$, and consider the system described by the transfer function $H_{yu}(s; h)$ in (1) in closed-loop with $K(s)$ given in (4). Then, the corresponding crossing points $\vec{k}_\sigma = (k_p(\omega, \sigma), k_i(\omega, \sigma)) \in \mathbb{R}^2$ are given by:*

$$k_p(\omega, \sigma) = \frac{e^{-h\sigma} (-\omega\alpha_{1,\sigma} \cos(h\omega) - \alpha_{2,\sigma} \sin(h\omega))}{\omega((z + \sigma)^2 + \omega^2)}, \quad (9)$$

$$k_i(\omega, \sigma) = \frac{e^{-h\sigma} (\sigma^2 + \omega^2) (\alpha_{3,\sigma} \cos(h\omega) - \alpha_{4,\sigma} \sin(h\omega))}{\omega((z + \sigma)^2 + \omega^2)}, \quad (10)$$

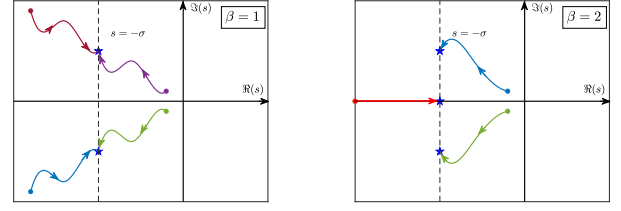
where:

$$\begin{aligned} d &= (\sigma + z)(\sigma(\sigma - a) + b), \\ \alpha_{1,\sigma} &= (\sigma(a\sigma + 2az - 2\sigma^2 - 3\sigma z) + \omega^2(a - 2\sigma + z) - bz), \\ \alpha_{2,\sigma} &= \omega^2(az + b - \sigma(a + 3z)) + \sigma d - \omega^4, \\ \alpha_{3,\sigma} &= \omega(\sigma^2 + 2\sigma z - az - b + \omega^2), \\ \alpha_{4,\sigma} &= d - \omega^2(a - \sigma + z). \end{aligned}$$

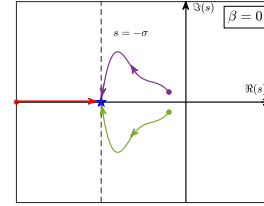
Moreover, when $\omega = 0$ the corresponding crossing points must satisfy:

$$(\sigma + z)k_i = \sigma(k_p(\sigma + z) - e^{-h\sigma}(\sigma(\sigma - a) + b)). \quad (11)$$

With this in mind, we define the σ -stability crossing curve $\mathcal{K}_\sigma = \mathcal{K}_{\omega,\sigma} \cup \mathcal{K}_{0,\sigma}$ in a similar manner as in Corollary 1, and we denoted the σ -stable region, if any, as $\mathcal{S}_\sigma$



(a) Double complex conjugate root, corresponding to the case root plus simple real root, corresponding to the case $\beta = 1$. (b) Simple complex conjugate root, corresponding to the case $\beta = 2$.



(c) Triple real root, corresponding to the case $\beta = 0$.

Fig. 2

IV. SPECTRAL-ABSCISSA OPTIMIZATION

In the following, we aim to identify the pair of gains (k_p, k_i) inside the stability region such that the closed-loop system's spectral abscissa is placed as far left as possible. In this spirit, we consider three possible scenarios: (i) Open-loop stable systems¹; (ii) Open-loop marginally stable systems²; (iii) Open-loop unstable systems³. The need of separating the analysis in different cases comes from the geometrical properties of the σ -stable regions in each scenario. We will identify the three different possible cases using $\beta \in \{0, 1, 2\}$, illustrated in Fig. 2, where (i) $\beta = 0$ corresponds to a spectral abscissa in a triple real root, (ii) $\beta = 1$ corresponds to a spectral abscissa in a double complex root and (iii) $\beta = 2$ correspond to a spectral abscissa in a simple complex root and a simple real root.

A. Open-loop stable system

We have the following results, corresponding to $\beta = 0$:

Proposition 4. *Consider the system described by the transfer function $H_{yu}(s; h)$ in (1) in closed-loop with the PI controller in (4), then, if all the roots of $Q(s)$ in (3) are in $\mathbb{C}_-$, then the region $\mathcal{S} \neq \emptyset$.*

Sketch of proof. The proof follows from continuity arguments. $\square$

Theorem 1 (Optimal σ in the open-loop stable systems case). *Consider the system described by the transfer function $H_{yu}(s; h)$ in (1) in closed-loop with the PI controller in (4),*

¹both roots of the polynomial $Q(s)$ in (3) are in $\mathbb{C}_-$

²both roots of the polynomial $Q(s)$ in (3) are in $i\mathbb{R}^*$

³at least one root of the polynomial $Q(s)$ in (3) is in $\mathbb{C}_+$

then the spectral abscissa $\rho(\Delta(s; h))$ achieves its minimum value $\bar{\sigma}$ at the smallest real positive root of

$$p(\sigma) := \sigma^5 + a_4\sigma^4 + a_3\sigma^3 + a_2\sigma^2 + a_1\sigma + a_0,$$

where:

$$\begin{aligned} a_4 &= 2z - a - \frac{4}{h} \\ a_3 &= z(z - 2a) + b + \frac{2}{h^2} - \frac{2(a - 5z)}{h} \\ a_2 &= \frac{z(-hz(ah + 6) + 6ah + 2bh^2 + 6)}{h^2} \\ a_1 &= \frac{z(2z(2ah + 3) + bh(hz - 2))}{h^2} \\ a_0 &= -\frac{2z(az + bh + b)}{h^2} \end{aligned}$$

Furthermore, the point corresponds to a triple real root.

Sketch of proof. From a geometrical observation and the $\mathcal{D}$ -decomposition theory [26], the collapse of the σ -stable region must occur at a triple real root, and therefore $\Delta(-\sigma; h) = \Delta'(-\sigma; h) = \Delta''(-\sigma; h) = 0$ must hold, leading to the given solution. $\square$

Remark 2. Note that in the case where both roots of $Q(s)$ are negative we have $a > 0$ and $b > 0$, therefore, $a_0 < 0$ and since $p(\sigma)$ is monic, there would be at least two coefficients with different signs, and from the Descartes' rule [27] we will have at least one real positive root.

Corollary 2 (Optimal control gains for open-loop stable systems). Consider the system described by the transfer function H_{yu} in (1) in closed-loop with the PI controller in (4), then the pair of gains that place the spectral abscissa of the system at $\bar{\sigma}$ are given by:

$$\begin{aligned} \bar{k}_p &= \frac{e^{-h\bar{\sigma}}}{(\bar{\sigma} + z)^2} (-\bar{\sigma}^2(-z(ah + 3) + a + bh) - z\bar{\sigma}(2a + bh) \\ &\quad + \bar{\sigma}^3(ah - hz + 2) + bz - h\bar{\sigma}^4), \\ \bar{k}_i &= \frac{\bar{\sigma}^2 e^{-h\bar{\sigma}}}{(\bar{\sigma} + z)^2} (\bar{\sigma}(\bar{\sigma} + h\bar{\sigma}(a - \bar{\sigma}) + z(ah - h\bar{\sigma} + 2)) \\ &\quad - az - b(h(\bar{\sigma} + z) + 1)). \end{aligned}$$

B. Open-loop unstable and marginally stable systems

In the case of open-loop unstable and marginally stable systems, the corresponding minimal spectral abscissa is either of the form $\beta = 1$ or $\beta = 2$. To find the corresponding $\bar{\sigma}$ and β we have the following result:

Proposition 5. Consider the system described by (1) in closed-loop with the PI controller $K(s)$ in (4). Assume that the optimal spectral abscissa $\bar{\sigma} \in [\sigma^{(-)}, \sigma^{(+)}] =: \Gamma \subset \mathbb{R}_+$ for appropriated $\sigma^{(-)}, \sigma^{(+)} \in \mathbb{R}_+$. For any $\sigma \in \Gamma$, let us denote by $\bar{\omega}_\sigma$ and $\underline{\omega}_\sigma$ such that $\underline{\omega}_\sigma \leq \bar{\omega}_\sigma$ and $\mathcal{K}_{\omega_\sigma, \sigma} = \mathcal{K}_{\bar{\omega}_\sigma, \sigma}$. Assume that for a given $\sigma \in \Gamma$ we have that $\text{card}\{\mathcal{S}_\sigma\} = 1$. Then, $\sigma = \bar{\sigma}$. Furthermore, if $\underline{\omega}_\sigma = \bar{\omega}_\sigma$. Such a $\bar{\sigma}$ can be computed by means of the following algorithm:

Algorithm 1 Optimization of the spectral abscissa

Let $\mathcal{S}$ be the stable region of (5). Set $j := 0$ and let (σ_j) be an increasing sequence in Γ such that $\sigma_0 := \sigma^{(-)}$ and $\sigma_\ell := \sigma^{(+)}$. Let $\mathcal{K}_{\sigma_j, \omega}$ such that $\omega \in \Omega_{\sigma_j} := [\underline{\omega}_{\sigma_j}, \bar{\omega}_{\sigma_j}]$.

- Step 1)** Let $N_j^{(s)} := \text{card}\{\mathcal{S}_{\sigma_j}\}$, and $N_j^{(k)} := \text{card}\{\mathcal{K}_{\omega, \sigma_j} \cap \mathcal{K}_{0, \sigma_j}\}$;
 - Step 2)** If $N_j^{(s)} \equiv 1 \wedge N_j^{(k)} \equiv 0$, then $\bar{\sigma} \equiv \sigma_j$, $\beta := 1$, if $N_j^{(s)} \equiv 1 \wedge N_j^{(k)} \equiv 1$, $\beta := 2$ and go to **Step 5**. Otherwise, go to **Step 3**;
 - Step 3)** Set $j := j + 1$;
 - Step 4)** If $j < \ell$ go to **Step 1**, otherwise, increase $\text{card}\{\Gamma\}$ and repeat the algorithm; go to **Step 5**;
 - Step 5)** End.
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It is worthy of mention that the stable region $\mathcal{S}$ does not always exist, that is, open-loop unstable and marginally stable systems cannot always be stabilized using a controller of the form (4). However, under the assumption that $\mathcal{S}$ exists, Proposition 5 and Algorithm 1 allow to find the optimal gains.

V. NUMERICAL EXAMPLES

The following examples aim to show the effectiveness of the presented results. All the numerical computations of quasipolynomial roots are computed using the QPmR algorithm in Matlab [28].

Example 1 (Open-loop stable system). Consider the system described by the following transfer function:

$$H_{yu}(s; h) = \frac{s - 1}{s^2 + s + 1} e^{-s}. \quad (12)$$

From Theorem 1 and Corollary 2 the system correspond to the case $\beta = 0$, and the minimum achievable spectral abscissa of the closed-loop system is given by $s = -\bar{\sigma} = -0.56$, and can be achieved by tuning the control parameters as $(\bar{k}_p = 0.049, \bar{k}_i = -0.12)$. The stability region of the closed-loop system is depicted in Fig. 3 (upper). The effect of placing the spectral abscissa as to the left as possible can be observed in Fig. 3 (lower) where we compare the closed-loop response of the system with the optimal pair of gains and two other pair of parameters inside the stable region.

Example 2 (Open-loop marginally stable system). Next we consider the case of a marginally stable system, which corresponds to an oscillator, with the transfer function given by:

$$H_{yu}(s; h) = \frac{s - 1}{s^2 + 4} e^{-s}. \quad (13)$$

From Proposition 5 the minimal spectral abscissa is of the form $\beta = 1$, and $\sigma = 1$. The corresponding $(\bar{k}_p = -0.23, \bar{k}_i = -1.25)$, placing the spectral abscissa at a double complex root at $s = -1 \pm 1.3i$. Fig. 4 depicts the location of the optimal gains (upper), as well as a closed-loop response comparison between the optimal control and two arbitrary stabilize controllers (lower).

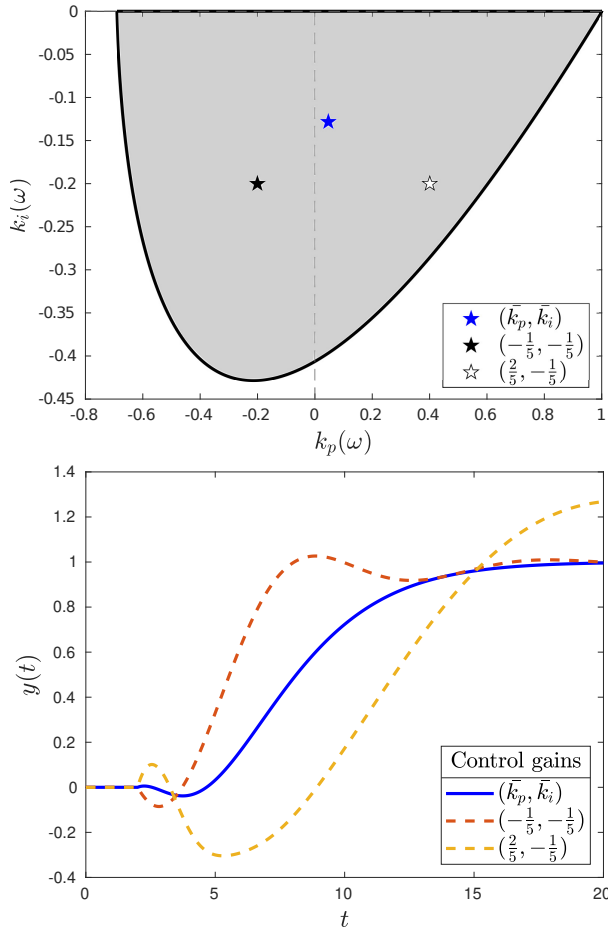


Fig. 3: (Upper) Stability region (gray) and optimal gains location for system (12) in closed-loop with a controller $K(s)$ of the form given by (4). (Lower) Closed-loop system time response comparison between the optimal control gains and two other arbitrary stabilizing controllers.

Example 3 (Open-loop unstable system). Consider the open-loop unstable system retrieved from [29], and described by the following transfer function:

$$H_{yu}(s; h) = \frac{s - 2}{s^2 - \frac{1}{2}s + \frac{13}{4}} e^{-2s}. \quad (14)$$

In this case, from Proposition 5 and Algorithm 1 the minimal spectral abscissa is of the form $\beta = 2$, and it corresponds to $\bar{\sigma} = 0.18$. The optimal gains are given by $(\bar{k}_p = -0.42, \bar{k}_i = -0.27)$, placing the spectral abscissa of the system at $s_{1,2} = \{-0.18, -0.18 \pm 2i\}$. A closed-loop comparison between the optimal gains and two other arbitrary controllers inside the stable region $\mathcal{S}$ is depicted in Fig. 5.

VI. CONCLUSIONS AND FUTURE WORK

In this note the problem of finding optimal control gains such that a second-order non-minimum phase time-delay systems achieves its minimal spectral abscissa was tackled. The effectiveness of the proposed results was showed to numerical examples.

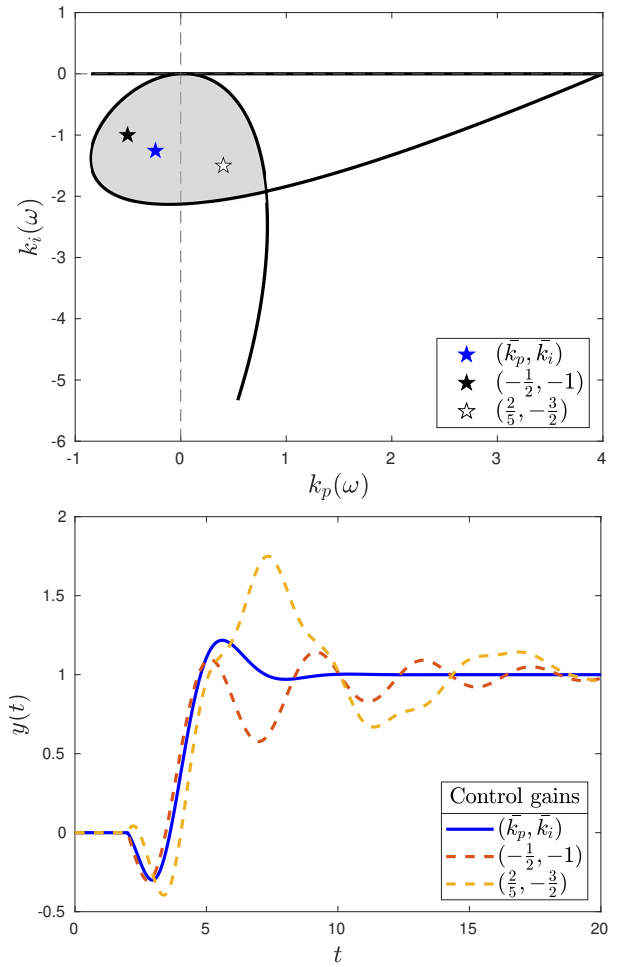


Fig. 4: (Upper) Stability region (gray) and optimal gains location for system (13) in closed-loop with a controller $K(s)$ of the form given by (4). (Lower) Closed-loop system time response comparison between the optimal control gains and two other arbitrary stabilizing controllers.

The problem of finding conditions under which there exist at least one stable region in the open-loop unstable case, as well as finding analytical expressions for the optimal gains in both the marginally stable and the unstable cases.

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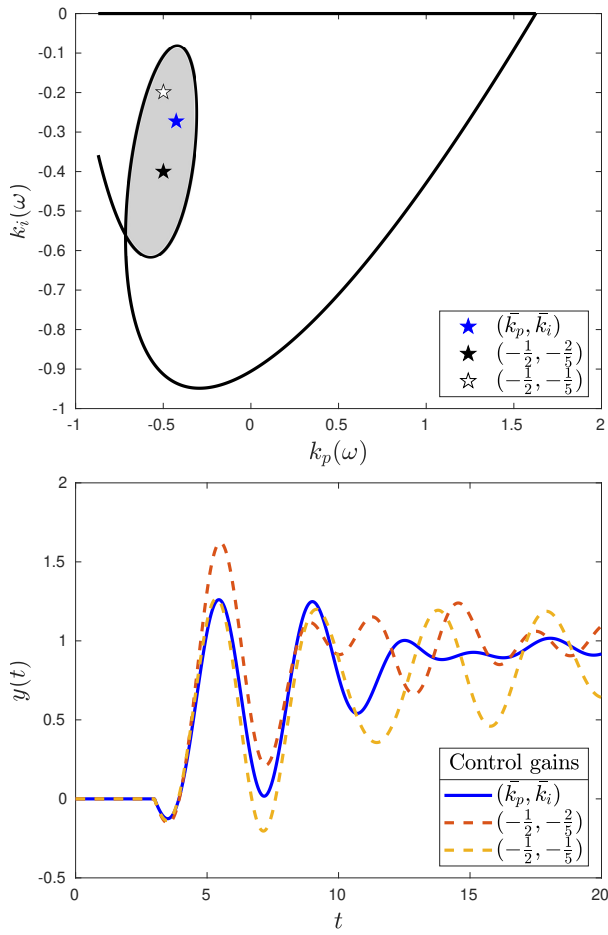


Fig. 5: (Upper) Stability region (gray) and optimal gains location for system (14) in closed-loop with a controller $K(s)$ of the form given by (4). (Lower) Closed-loop system time response comparison between the optimal control gains and two other arbitrary stabilizing controllers.

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